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Project-2: Motor Winding Design and Analysis

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Introduction

Integral Slot Winding Design

In this section, the winding design is conducted for two different integer-slot winding 72 slots, 6 pole and 3 phase machines, one for the full-pitched winding case and one for 11/12 short pitched case.

Full-Pitched Winding

In this section, the winding of 72 slot, 6 pole, 3 phase machine designed with full-pitched coils.

For the given configuration, we first calculate the number of slots per pole per phase.

$$Q = 72 \text{ slots} \quad (1)$$

$$p = 6 \text{ poles} \quad (2)$$

$$m = 3 \text{ phases} \quad (3)$$

$$q = \frac{Q}{pm} = \frac{72}{6 * 3} = 4 \text{ slots per pole per phase} \quad (4)$$

Also, we can find the electrical slot angle as,

$$\alpha_u = \frac{360^\circ}{Q/(p/2)} = \frac{360^\circ}{72/3} = 15^\circ \quad (5)$$

So, each successive slot has a relative phase angle of 15° . If we denote the slot index by i , then, slot i has a phase angle of $(i - 1) * 15^\circ$. Also, we know that each pole occupies 4 slots. If we place phase A_{1+} to slot $i = 1$ (0 degrees), then we should place phase A_{1-} at an electrical phase of 180° , since we have a full-pitch winding. Since the electrical slot angle is 15° , we should place the return line A_{1-} at slot $i = (180/15) = 13$. With the same logic, we can find the slot placement of each of the phases. The winding diagram in this case is given in 1.

Winding Diagram Layout

Slot	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Angle	0°	15°	30°	45°	60°	75°	90°	105°	120°	135°	150°	165°	180°	195°	210°	225°	240°	255°	270°	285°	300°	315°	330°	345°
	A_{1+}	A_{2+}	A_{3+}	A_{4+}	C_{5-}	C_{6-}	C_{7-}	C_{8-}	B_{1+}	B_{2+}	B_{3+}	B_{4+}	A_{5-}	A_{6-}	A_{7-}	A_{8-}	C_{1+}	C_{2+}	C_{3+}	C_{4+}	B_{5-}	B_{6-}	B_{7-}	B_{8-}
	A_{5+}	A_{6+}	A_{7+}	A_{8+}	C_{1-}	C_{2-}	C_{3-}	C_{4-}	B_{5+}	B_{6+}	B_{7+}	B_{8+}	A_{1-}	A_{2-}	A_{3-}	A_{4-}	C_{5+}	C_{6+}	C_{7+}	C_{8+}	B_{1-}	B_{2-}	B_{3-}	B_{4-}

Table 1: Winding Diagram for the Full-Pitched Integer Slot Winding Design

Distribution Factor, Pitch Factor and Winding Factor

The distribution factor for n^{th} harmonic can be calculated as follows.

$$k_d(n) = \frac{\sin(qn \frac{\alpha_u}{2})}{q \sin(n \frac{\alpha_u}{2})} \quad (6)$$

Also, the pitch factor for n^{th} harmonic can be calculated as follows.

$$k_p(n) = \sin(n \frac{\lambda}{2}) \quad (7)$$

In these equations, n is the harmonic number and λ is the coil pitch. In the full pitch case, we have $k_p = 1$ for all the harmonics since $\lambda = 180^\circ$. The distribution factor for the fundamental component, 3^{rd} and 5^{th} components can be calculated as follows.

$$k_d(1) = \frac{\sin(4 * \frac{20^\circ}{2})}{4 \sin(\frac{20^\circ}{2})} = 0.925 \quad (8)$$

$$k_d(3) = \frac{\sin(4 * 3 * \frac{20^\circ}{2})}{4 * \sin(3 \frac{\alpha_u}{2})} = 0.433 \quad (9)$$

$$k_d(5) = \frac{\sin(4 * 5 * \frac{20^\circ}{2})}{4 \sin(5 * \frac{20^\circ}{2})} = -0.112 \quad (10)$$

Thus, the winding factor $k_w = k_d * k_p$ for the fundamental, 3^{rd} and 5^{th} components can be calculated as follows.

$$k_w(1) = 0.925 \quad (11)$$

$$k_w(3) = 0.433 \quad (12)$$

$$k_w(5) = -0.112 \quad (13)$$

Comments

From the winding factor, the distributed coil causes to drop the effect of 3^{rd} harmonic to 43.3% of its magnitude and 5^{th} harmonic to 11.2% with a phase difference of 180° due to the minus sign. The tradeoff is that the fundamental component magnitude drops to its 92.5% due to the effect of the distribution. In this case, we have no effect of the coil pitch on the winding factor since the coil is full-pitch. The winding factor can be better (lower for the harmonics) with the introduction of the short-pitch winding.

11/12 Short-Pitched Winding

For the short-pitch case, the slot angle and the number of slots per pole per phase is the same as the full-pitch case. The only difference is the coil pitch, which can be calculated as,

$$\lambda = \frac{11}{12} * 180^\circ = 165^\circ \quad (14)$$

So, the return line of each coil is phase shifted 165° electrically. We know the slot angle is 15° , so, if we place phase A_{1+} to slot $i = 1$, then we should place the return line A_{1-} to slot $i = (165/15) + 1 = 12$. With a similar logic for other phases, we obtain the winding diagram given in 2.

Winding Diagram Layout

Slot	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Angle	0°	15°	30°	45°	60°	75°	90°	105°	120°	135°	150°	165°	180°	195°	210°	225°	240°	255°	270°	285°	300°	315°	330°	345°
	A_{1+}	A_{2+}	A_{3+}	A_{4+}	C_{5-}	C_{6-}	C_{7-}	C_{8-}	B_{1+}	B_{2+}	B_{3+}	B_{4+}	A_{5-}	A_{6-}	A_{7-}	A_{8-}	C_{1+}	C_{2+}	C_{3+}	C_{4+}	B_{5-}	B_{6-}	B_{7-}	B_{8-}
	A_{6+}	A_{7+}	A_{8+}	C_{1-}	C_{2-}	C_{3-}	C_{4-}	B_{5+}	B_{6+}	B_{7+}	B_{8+}	A_{1-}	A_{2-}	A_{3-}	A_{4-}	C_{5+}	C_{6+}	C_{7+}	C_{8+}	B_{1-}	B_{2-}	B_{3-}	B_{4-}	A_{8+}

Table 2: Winding Diagram for the 11/12 Short-Pitched Integer Slot Winding Design

Distribution Factor, Pitch Factor and Winding Factor

In this case, the distribution factor is the same as the full-pitch case since the number of slots per pole per phase is the same.

$$k_d(1) = \frac{\sin(4 * \frac{20^\circ}{2})}{4 \sin(\frac{20^\circ}{2})} = 0.925 \quad (15)$$

$$k_d(3) = \frac{\sin(4 * 3 * \frac{20^\circ}{2})}{4 * \sin(3 * \frac{20^\circ}{2})} = 0.433 \quad (16)$$

$$k_d(5) = \frac{\sin(4 * 5 * \frac{20^\circ}{2})}{4 \sin(5 * \frac{20^\circ}{2})} = -0.112 \quad (17)$$

The difference comes from the pitch factor.

$$k_p(1) = \sin(\frac{165^\circ}{2}) = 0.991 \quad (18)$$

$$k_p(3) = \sin(3 * \frac{165^\circ}{2}) = -0.924 \quad (19)$$

$$k_p(5) = \sin(5 * \frac{165^\circ}{2}) = 0.793 \quad (20)$$

Hence, the winding factors for the fundamental, 3rd and 5th components become,

$$k_w(1) = 0.925 * 0.991 = 0.917 \quad (21)$$

$$k_w(3) = 0.433 * (-0.924) = -0.400 \quad (22)$$

$$k_w(5) = (-0.112) * 0.793 = -0.089 \quad (23)$$

Comments

In this case, utilizing the short-pitch winding, we obtain a better performance in terms of the harmonics rejection, with a very small loss in the fundamental component. Also, with the short pitch winding, a phase difference of 180° is added on the third harmonic.

Fractional-Slot Winding Design for Reference Design 2

Phase Angle of Induced Voltage in Each Slot

Phasor Diagram for One Phase and Winding Factor

Complete Winding Diagram

Slot	1		2		3		4		5		6		7		8		9		10		11		12		13		14		15	
Angle	0°		120°		240°		0°		120°		240°		0°		120°		240°		0°		120°		240°		0°		120°		240°	
	B_{5-}	A_{1+}	A_{1-}	C_{1+}	C_{1-}	B_{1+}	B_{1-}	A_{2+}	A_{2-}	C_{2+}	C_{2-}	B_{2+}	B_{2-}	A_{3+}	A_{3-}	C_{3+}	C_{3-}	B_{3+}	B_{3-}	A_{4+}	A_{4-}	C_{4+}	C_{4-}	B_{4+}	B_{4-}	A_{5+}	A_{5-}	C_{5+}	C_{5-}	B_{5+}

Table 3: Winding Diagram for the Fractional-Slot Winding Design

Analytical Modelling

Electrical Loading

Carter's Coefficient and Effective Air-Gap Clearance

Effective Axial Length

Air-Gap Peak Magnetic Flux Density

Magnetic Loading, Specific Machine Constant, Shear Stress and Expected Torque

Flux Per Pole

2D FEA Modelling

Geometry

Meshing

Boundary Conditions

Solver Settings

Results

Flux Density Distribution

Air-Gap Flux Density Distribution

Leakage Flux Paths

Flux-Per-Pole

Bonus: Cogging Torque